

MultiCollinearity

b/w IV and
DV

Correlation $\rightarrow$ $\overset{\text{IV}}{X} - \overset{\text{target}}{Y} \approx 0.95$
 ≈ -0.95

interpretation

$\rightarrow$ if there is 1 unit increase in X , there is 95% chance that Y will also increase.

$\rightarrow$ $(X_1, X_2, X_3, \dots, X_{100})$ (all are IV)

$$X_1 \approx X_2$$

$$X_1 \approx (X_2, X_3, X_4, X_5, \dots)$$

$$X_3 \approx (X_1, X_2)$$

$$X_2 \approx (X_1, X_3, X_4)$$

Correlation b/w only two IVs at a time

$$\text{Corr}(X_1, X_2) = 0.98$$

$$\text{Corr}(X_3, X_4) = (0.97)$$

Two points

① $\rightarrow$ if X has a high correlation with $Y \rightarrow$ the model will be good.

What will happen if the features among themselves has a good correlation?

② $\rightarrow$ To measure relationship b/w two features we have Correlation. But how to measure correlation among multiple features?

① Correlation among features itself.

Problem

if $\text{Corr b/w } X_1, X_2 = 0.95$

↓
 $X_1 \approx X_2$ (in terms of variation,
 X_1 & X_2 is almost same)

if $X_1 = X_2$

$$\begin{cases} Y = \frac{8X_1 + 2X_2}{10X_1} \\ Y = 10X_2 \\ Y = 2X_1 + 8X_2 \end{cases}$$

the interpretation of MLR becomes

tough. (which feature is contributing more to Y will be tough to decide as $X_1 \approx X_2$.)

② Measure for Correlation among multiple features

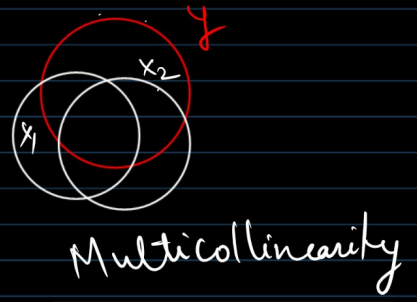
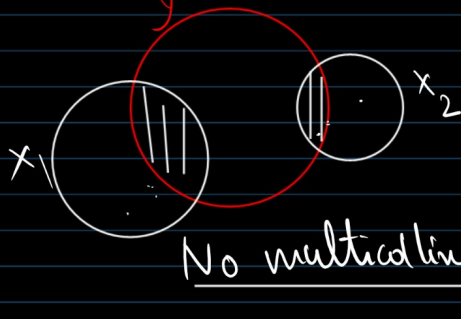
Multicollinearity

↓
multi — col — linearity
↓ ↓ ↓
many together linear relationship

Multi collinearity

Collinearity → when two features are linearly associated (high correlation) and they are used to predict target variable.

multi-collinearity → where a feature exhibits a linear relationship with more than two variables.



Why multicollinearity is a concern?

→ Affects interpretation

$$x_1 = x_2$$

$$8x_1 + 2x_2 = y$$

→ Increases Computation.

if $x_1 = x_2$, then why to use both the features in building model? It will increase computation.

→ It increases Overfitting

$$x_1 \approx y$$

$$x_1, x_2 \approx y$$

$\Rightarrow$

{ if both x_1 & x_2 is almost same, then pattern will be memorised, that's why it leads to overfitting.

Solution

① VIF (variation inflation factor) and drop feature one by one with high VIF.

② RFE (Recursive Feature elimination)

$x_1 - x$ (correlation / heatmap)

$$x_1 \approx (x_2, x_3, x_4, \dots) \rightarrow \underline{\underline{VIF}}$$

* VIF :- is a measure of amount of multicollinearity in regression.

$$\frac{x_1}{x_2, x_3}$$

$$x_1 \approx (x_2, x_3)$$

$$VIF_i = \frac{1}{1 - R_i^2}$$

(R^2 square - %age variation in y explained by x)

x_1, x_2, x_3, x_4, x_5

$$x_1 \approx (x_2, x_3, x_4, x_5)$$

$$VIF_{x_1} = \frac{1}{1 - R_{x_1}^2}$$

$$x_1 \rightarrow y$$

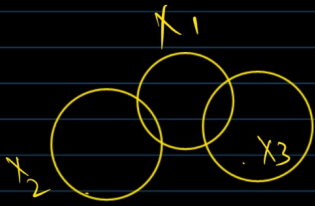
$$\underbrace{x_2, x_3, x_4, x_5} \rightarrow x_1$$

x_1, x_2, x_3

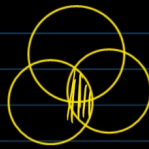
$x_1, x_2, x_3, y \rightarrow$ main model

$R_{square} = R^2 =$ % age variation explained in y by x

- Case-1: $x_1 \approx (x_2, x_3) \rightarrow VIF_{x_1} \rightarrow \frac{x_1}{y} \& \frac{x_2, x_3}{x} \Rightarrow R_{square}$
 Case-2: $x_2 \approx (x_1, x_3) \rightarrow VIF_{x_2} \rightarrow \frac{x_2}{y} \& \frac{x_1, x_3}{x} \Rightarrow \frac{1}{1 - R_{x_1}^2}$
 Case-3: $x_3 \approx (x_1, x_2), VIF_{x_3} = \frac{x_3}{y} \& \frac{x_1, x_2}{x} = \frac{1}{1 - R_{x_3}^2}$



No multi collinearity



low multicollinearity



High multicollinearity

$$\rightarrow VIF \geq 10$$

When features have $VIF > 10$, then we say high multicollinearity, then drop the features one by one.

VIF

$$10 = \frac{1}{1 - R_{square}}$$

$$1 - R_{square} = \frac{1}{10}$$

$$R_{sq} = 1 - \frac{1}{10} = \frac{10-1}{10} = \frac{9}{10} = \underline{\underline{0.90}}$$

$$X_1 \approx X_2 X_3 X_4$$

90% variance in X_1 is explained
by $X_2 X_3 X_4$

then why to keep X_1 .
You can drop X_1 (after
taking permission from
business team)

$$\begin{matrix} X_1 & X_2 & X_3 & X_4 \\ \sim & \sim & \sim & \sim \end{matrix}$$

Feature	VIF
X_1	12
X_2	13
X_3	8
X_4	7

$VIF \geq 10 \Rightarrow$ drop the features
one by one until
all the features
has $VIF < 10$.

$$X_1 \approx (X_2 X_3 X_4)$$

$$\rightarrow \underline{X_2} \approx (\underline{X_1} X_3 X_4) \quad (\text{say drop } X_2)$$

Next time
if you calculate
VIF $X_1 \approx (X_3 X_4)$

So X_2 will not
be there, so VIF
will change.

* Two kinds of multicollinearity

① Data-based collinearity $\rightarrow$ Present in
data itself.

latitude	longitude
{	}

② Structural multicollinearity

Caused due to new feature from
existing feature.

Distance	time	<u>Speed</u> = $\frac{\text{Distance}}{\text{time}}$
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* We talked that using VIF drop the feature one by one.

1000 features

RFE — Recursive feature Elimination.

- It will make a model with all 1000 features
- Start dropping one by one the least important features.
- Untill the desired no of feature is achieved.

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 → PCA
 → Regularisation (Ridge & lasso)